

# An Anisotropic Polynomial Dual Optimizer Beyond the Displayed Family

Candidate counterexample for arXiv:2008.07926v1

July 23, 2026

**Status.** Candidate counterexample, likely valid pending expert review. The target is the informal uniqueness belief following Example 2.20 in the arXiv v1 paper. The sentence is absent from the published JMAA version, and no published theorem is challenged.

## 1 Source statement

Gladkov, Kolesnikov, and Zimin study a  $(3, 2)$  multistochastic problem on  $[0, 1]^3$ . Its cost is  $c(x, y, z) = xyz$ . In Example 2.11, the primal measure is supported on the plane

$$\Gamma = \{(x, y, z) \in [0, 1]^3 : x + y + z = 1\},$$

and its three pair projections are prescribed. Example 2.20 displays a one-parameter family of symmetric dual optimizers

$$(f_A(x, y), f_A(x, z), f_A(y, z)), \quad A \geq 0,$$

and then states: “We believe that there are no other dual solution, but can not prove this” [1]. The relevant passage is on PDF page 8.

**Example 2.19. Example without dual solutions satisfying (1); (b) fails.** This example is considered in Theorem 6.32. In this example  $f_{xy}$  is either zero or not defined.

**Example 2.20. Non-uniqueness for the dual problem; (c) fails.** In the problem considered in Example 2.11 there exist many dual solutions. To see this let us note that the following inequality holds for all  $(x, y, z) \in [0, 1]^3$  and a fixed constant  $A > 0$ , equality holds if and only if  $x + y + z = 1$ :

$$(x + y + z - 1)^2(x + y + z + A) \geq 0.$$

Developing the left-hand side we see that this inequality is equivalent to

$$xyz \geq f_A(x, y) + f_A(x, z) + f_A(y, z),$$

where

$$f_A(x, y) = -\frac{1}{12}(x^3 + y^3) - \frac{1}{2}xy(x + y) - (A - 2)\left(\frac{x^2}{12} + \frac{xy}{3} + \frac{y^2}{12}\right) - \frac{1 - 2A}{12}(x + y) - \frac{A}{18}.$$

Clearly, the triple  $(f_A(x, y), f_A(x, z), f_A(y, z))$  solves the dual problem for every  $A \geq 0$ . Note that (1) and (c) fails for all  $A \geq 0$ .

We believe that there are no other dual solution, but can not prove this.

The journal version of the paper contains the following sentence: “We believe that there are no other dual solution, but can not prove this.”

The journal version retains Example 2.20 but removes the quoted sentence. Accordingly, the result below is scoped to the literal arXiv-version belief.

## 2 Setup

For reference, the source polynomial is

$$f_A(u, v) = -\frac{u^3 + v^3}{12} - \frac{1}{2}uv(u + v) - (A - 2)\left(\frac{u^2}{12} + \frac{uv}{3} + \frac{v^2}{12}\right) - \frac{1 - 2A}{12}(u + v) - \frac{A}{18}.$$

Writing  $s = x + y + z$ , direct expansion gives

$$xyz - f_A(x, y) - f_A(x, z) - f_A(y, z) = \frac{(s - 1)^2(s + A)}{6}. \quad (1)$$

This is nonnegative on the cube and vanishes on  $\Gamma$ , proving feasibility and complementary optimality of the source family.

## 3 Proof intuition

The square  $(s - 1)^2$  is exactly what forces equality on the support plane. The source multiplies it by the symmetric affine function  $s + A$ . To leave the class of pairwise potentials, a perturbation would need a genuine three-variable interaction. But

$$(s - 1)^2(x - y)$$

has no  $xyz$  monomial: its mixed third derivative is zero. It therefore splits as a sum of three pair polynomials. A small multiple can be subtracted from the positive source gap without changing its sign. This produces an anisotropic optimal triple whose total sum cannot be obtained from the symmetric family by either changing  $A$  or applying the usual gauge freedom.

## 4 Counterexample

**Theorem 1.** *For the planar multistochastic problem of Examples 2.11 and 2.20, define*

$$\begin{aligned} h_{12}(x, y) &= (x - y)(x + y - 1)^2, \\ h_{13}(x, z) &= xz(2x + z - 2), \\ h_{23}(y, z) &= -yz(2y + z - 2), \end{aligned}$$

and set

$$\begin{aligned} g_{12}(x, y) &= f_1(x, y) + \frac{1}{12}h_{12}(x, y), \\ g_{13}(x, z) &= f_1(x, z) + \frac{1}{12}h_{13}(x, z), \\ g_{23}(y, z) &= f_1(y, z) + \frac{1}{12}h_{23}(y, z). \end{aligned}$$

Then  $(g_{12}, g_{13}, g_{23})$  is a polynomial dual optimizer. It is not gauge-equivalent to the source triple for any parameter  $A \geq 0$ .

*Proof.* First, exact expansion gives

$$h_{12}(x, y) + h_{13}(x, z) + h_{23}(y, z) = (s - 1)^2(x - y). \quad (2)$$

Combining (1) at  $A = 1$  with (2) yields

$$\begin{aligned} xyz - g_{12}(x, y) - g_{13}(x, z) - g_{23}(y, z) &= \frac{(s-1)^2(1+s)}{6} - \frac{(s-1)^2(x-y)}{12} \\ &= \frac{(s-1)^2(2+x+3y+2z)}{12}. \end{aligned} \tag{3}$$

Every factor on the last line is nonnegative on  $[0, 1]^3$ . Thus the new triple is dual feasible. On  $\Gamma$  the right side vanishes, so the dual sum equals  $xyz$  on the support of the primal measure. Integrating against that measure shows that the dual value equals the primal value. Hence the new triple is optimal.

It remains to prove that this is not a disguised member of the displayed family. Gauge transformations of pair potentials have identically zero total sum, so gauge-equivalent triples must have the same total sum at every  $(x, y, z)$ . If the new total were the total associated with  $f_B$  for some  $B$ , comparison of (3) and (1) would require

$$2 + x + 3y + 2z = 2(s + B)$$

identically. This is impossible: the coefficients of  $x$  and  $y$  on the left are 1 and 3, whereas both coefficients on the right are 2.  $\square$

## 5 Verification and scope

The identities were checked over the exact rational polynomial ring by the included SymPy script. No numerical approximation enters the proof. The sign in (3) is coefficientwise transparent, and complementary optimality uses only the source’s planar support statement.

The construction disproves exhaustiveness of the displayed symmetric one-parameter family; it does not classify all optimizers. It also does not answer the paper’s broader Open Problem 2 about sufficient conditions for an optimal-map representation.

## 6 Novelty check

The run indexes and bounded web searches on July 23, 2026 used arXiv:2008.07926, the exact quoted sentence, the authors, Example 2.20, the notation  $f_A$ , “dual uniqueness,” and the perturbation  $(x + y + z - 1)^2(x - y)$ . No explicit later answer or matching construction was found. Novelty confidence is moderate-to-low because the published 2022 version silently removes the uniqueness sentence.

## 7 Human review recommendation

The decisive checks are the decomposition (2), the exact gap (3), and the scope of the informal arXiv-only statement. The first two are elementary polynomial identities; the last determines how the result should be described historically.

## References

- [1] N. A. Gladkov, A. V. Kolesnikov, and A. P. Zimin, *The multistochastic Monge–Kantorovich problem*, arXiv:2008.07926v1 (2020); J. Math. Anal. Appl. **506** (2022), 125666.